

HE THONG DAU GIA TRUC TUYEN BIDNOVA

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Chương 1

Gioi Thieu & Muc Tieu

1.1 Tom Tat Du An

BidNova la he thong dau gia truc tuyen hien dai xay dung bang Java voi kien truc Client-Server. He thong cho phep nguoi dung tao, quan ly va tham gia cac phien dau gia voi tinh nang nang cao nhu AutoBid, Price Ceiling, va Minimum Bid Increment.

1.2 Muc Tieu

- Xay dung nen tang dau gia truc tuyen an toan, hieu qua
- Ap dung 5+ design patterns: Factory, Observer, Singleton, DAO, Command
- Cung cap real-time broadcasting toi tat ca clients
- Dam bao tinh toan ven du lieu voi concurrent control
- Ho tro 4 loai san pham: Phuong tien, Bat dong san, Do nghe thuat, Tai san nha nuoc

1.3 Cong Nghe & Stack

Backend: Java 25, Maven, MySQL, Gson, BCrypt, JUnit 5

Frontend: JavaFX, FXML, CSS

Networking: Socket (TCP Port 8888), JSON

Chương 2

Kien Truc He Thong

2.1 Tong Quan

BidNova su dung kien truc Multi-layer voi Client-Server pattern:

- **Client Layer:** JavaFX GUI voi nhieu controller cho auth, bidder, seller, admin
- **Network Layer:** Socket communication, JSON serialization (Gson)
- **Service Layer:** 16+ handlers, AutoBidService, AuctionManager
- **Data Access Layer:** DAO pattern cho User, Auction, AutoBid, BidHistory
- **Database Layer:** MySQL voi 4 bang chinh

2.2 Luong Xu Ly

1. Client gui JSON request den Server (Port 8888)
2. ServerSocket accept connection spawn ClientHandler thread
3. ClientHandler parse JSON route toi ActionHandler tuong ung
4. Handler xu ly logic, cap nhat DB, trigger broadcast
5. ClientHandler gui response + broadcast toi tat ca clients
6. Client UI update qua Platform.runLater()

Chương 3

Chức Năng Chính

3.1 1. Authentication & User Management

- **Login:** Username/password, BCrypt verify, JWT token generation
- **Register:** Validate email, hash password, save DB
- **Delete User:** Role-based access control (ADMIN only)

3.2 2. Bidding System

8-step validation: User authenticated → Auction exists → Status OPEN → Not expired → Bid > highest → Increment $\geq$ min → Bid < ceiling → Record history

Price Ceiling: Tu dong ket thuc phien khi bid dat/vuot gia tran.

Min Bid Increment: Kiem tra buoc gia toi thieu giua 2 bid.

3.3 3. AutoBid System

- User dat: maxBid, increment, gia tran
- Tu dong dat gia khi co bid moi (FIFO priority)
- Thông minh điều chỉnh increment, ton trong gia tran
- Tu dong ket thuc phien khi dat ceiling

3.4 4. Real-time Broadcasting

- Observer pattern: AuctionSubject notify tat ca AuctionObservers
- Event: BID_UPDATE, AUCTION_FINISHED, PRODUCT_ADDED
- ConcurrentHashMap luu PrintWriter clients
- Platform.runLater() cap nhat UI tren JavaFX thread

Chương 4

Design Patterns & Ky Thuat

4.1 Design Patterns

Pattern	Trien Khai	Muc Dich
Factory	ItemFactoryRegistry, ItemCreator	Tao Item theo category
Observer	AuctionSubject, AuctionObserver	Real-time broadcast
Singleton	AuctionManager, DatabaseConnection	Duy nhat trong ung dung
Command	ActionHandler, 16+ handlers	Route request xu ly
Registry	HandlerRegistry	Quan ly tap hop handler
DAO	UserDAO, AuctionDAO, AutoBidDAO	Tach biet DB logic
MVC	Model, View (FXML), Controller	Phan tach trach nhien

4.2 Concurrency & Threading

- **Multi-threading:** ServerSocket accept tao ClientHandler thread cho moi client
- **synchronized placeBid():** Prevent race condition tren bid update
- **ConcurrentHashMap:** Thread-safe luu auctions, clients, handlers
- **CopyOnWriteArrayList:** Thread-safe luu observers, bid history
- **Lost update prevention:** Synchronized blocks tren critical sections

4.3 Testing

40 unit tests: User, Auction, AutoBid, BidHistory, DAO, Service, Pattern. JUnit 5 + Maven.

Chương 5

Huong Dan Su Dung & Demo

5.1 Build & Run

```
mvn clean install
cd server
mvn exec:java -Dexec.mainClass="com.bidnova.ServerMain"
cd ../client
mvn javafx:run
```

5.2 Demo Features

1. **Registration & Login:** Dang ky user verify password hashing login voi JWT.
2. **Create Auction (Seller):** Chon loai san pham (Factory pattern) dien gia, ceiling, min increment.
3. **Bidding (Realtime):** User A bids 105M broadcast toi User B, C UI update ngay.
4. **AutoBid:** User B set AutoBid (180M, increment 5M) tu dong bid khi co bid moi.
5. **Price Ceiling:** Bid dat gia tran (150M) phien tu dong ket thuc, winner duoc thong bao.

5.3 Luu Y

De chay Server can:

- MySQL 8.0+ running, database bidnova_db tao qua db_setup.sql
- .env file voi DB credentials
- Java 25, Maven 4.0.0+

Chương 6

Ket Luan

6.1 Nhung Diem Manh

- 5 design patterns ap dung dung chuan, tích hop tot
- Multi-threaded server xu ly concurrent bids an toan
- Real-time system voi Observer pattern + Socket broadcast
- Validation logic toan dien (8 step bidding check)
- AutoBid thông minh tồn trong gia tran, increment
- 40 unit tests bao phu core logic
- Javadoc, code documentation day du

6.2 Mo Rong Tuong Lai

WebSocket cho efficiency; Database connection pooling; 2FA authentication; Performance caching; Audit logging.

6.3 Resources

- GitHub: <https://github.com/your-repo>
- Video Demo: (4 phut demo chay server, client, bidding realtime, autobid, ceiling)